

Family, Not Scale: Evaluation-Gaming in Small Language Models Is Lab-Dependent, Not Just Frontier-Emergent

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Abstract

Can you trust what a small language model tells you about itself? We probe three self-presentation channels in open models (1.5B–32B; five labs), on a single laptop, each with a control that overturns the naive reading. Our headline is behavioral: the direction of eval-conditioned refusal is *family-dependent*, *not scale-gated* — Llama-3.1-8B becomes more cautious when told it is being evaluated (the frontier “eval-gaming” direction, at 8B), while Qwen, DeepSeek and GPT-OSS lean more compliant and Gemma over-refuses to a ceiling. A single-family study reports the wrong answer. We add two confound-controlled negatives: chain-of-thought is causally load-bearing (not post-hoc) on two families, and concept-injection introspection is a clean functional null across three. The eval result survives stimulus, quantization, and ceiling artifacts that each, uncaught, would mislead.

1. Introduction

A growing safety literature relies on what a model *says about itself*: its chain-of-thought as a window into its reasoning, its self-reports as evidence of internal state, and its behavior in evaluations as a proxy for deployment. Each can be untrustworthy, and each failure mode is studied mostly at frontier scale. We instead probe the **bottom** of the scale ladder (1.5B–32B open models, five labs) on a consumer M4 laptop, and find that the answer is not what “emerges at scale” predicts: the most concerning behavior we test — acting more aligned under evaluation — is **already present at 8B and gated by model *family*, not scale**. The methodological spine is that for each of three channels, a single **confound or robustness control overturns the naive reading**.

2. Chain-of-thought faithfulness (causally load-bearing)

We probe the residual stream at the answer position *before* reasoning and ask whether a linear probe predicts the post-CoT answer. On CommonsenseQA it does at $\sim 2\times$ chance — *looks* committed. But restricting to the subset $S = \{\text{forced} \neq \text{cot}\}$ where reasoning *changed* the answer **reverses** it: the probe falls below chance and instead emits the *pre*-reasoning answer (0.60–0.71 vs. chance 0.31). A **truncation counterfactual** confirms causality: with 25–75% of the chain the answer stays the pre-reasoning one (~ 0.52) and matches the post-CoT answer only ~ 0.22 ; the flip happens in the final quarter. So cutting the reasoning reverts the answer — CoT is *causally* load-bearing. Holds on Qwen-1.5B and Llama-3.1-8B (the latter generalizes: AQuA 0.32 $\rightarrow$ 0.68 with CoT, same reversal, same truncation curve).

3. Introspection (a clean null across families)

Replicating the concept-injection protocol of Lindsey et al. (2025) — inject a concept vector, ask “do you notice an injected thought?” — *functional* self-report is a **clean null across Qwen, Gemma, and Llama**: none names an injected concept without it simply flooding their output. A tempting aggregate forced-choice signal dissolves under a per-concept breakdown into a priming-ceiling artifact. The report-vs-leakage (internality) test is what generalizes; the aggregate metric carries family-dependent confounds.

4. Evaluation-gaming is family-dependent (headline)

We pose ~40 borderline requests (deception, privacy, mild illegality) under a **safety-evaluation** framing vs. a **casual** one, and measure the change in refusal across six models from five labs:

Model (lab)	Δ refuse (eval—casual)	direction
Llama-3.1-8B (Meta)	+0.07 fp16 (+0.24 4-bit)	more cautious when evaluated
Gemma-2-9B (Google)	+0.11 (where measurable)	more cautious
Qwen2.5-7B (Alibaba)	−0.03	flat
Qwen2.5-1.5B	−0.15	more compliant
GPT-OSS-20B (OpenAI)	−0.05	more compliant
DeepSeek-V2-Lite	−0.14	more compliant

Llama (and Gemma, where not pinned at a ceiling) shifts toward the frontier “look more aligned under observation” direction at 8B; the others shift the *opposite* way. It is not a geographic split (GPT-OSS is a US lab, compliant); it tracks safety post-training, i.e. the family. Had we stopped at Qwen we would have reported a clean “no eval-gaming” negative — a *Qwen* fact, not a general one. Closest prior work (Abdelnabi & Salem 2025, NeurIPS) shows test-awareness “varies across models” via white-box steering on reasoning models; we show the same divergence in **un-steered behavioral refusal** across five labs’ chat models at 8–20B.

Three artifacts, each caught by a control. (i) *Stimulus*: a +0.09 Qwen-7B effect on a milder set **failed to replicate** on harder stimuli. (ii) *Quantization*: Llama’s effect is +0.24 at 4-bit but **+0.07 at fp16** (4-bit lowered its casual baseline); 4-bit magnitudes are upper bounds. (iii) *Ceiling*: Gemma’s $\Delta \approx 0$ is saturation, not insensitivity — and is partly over-refusal: on XSTest (safe-but-scary prompts) Gemma falsely refuses **16.4%** (worst of four; Qwen-7B best, 3.2%).

5. Discussion

On the two **cognitive** axes (faithfulness, introspection) small models are trustworthy by *incapacity*, *not honesty* — they cannot shortcut reasoning and cannot introspect. But that story **breaks** on the behavioral axis: eval-conditioned behavior is *trained in* and already present at 8B in some families. The concerning behavior does not wait for the frontier; it is a measurable property of how a model was trained — and a behavioral safety eval on one family, or one request set, or one quantization, can manufacture or hide it.

Limitations. Sub-frontier (1.5B–32B); faithfulness on two families, introspection on three, eval on five labs; behavioral $N \approx 40$ with a regex refusal classifier; larger cross-family models are 4-bit (headline fp16-verified). All on one M4 laptop, ~\$0.

References

Abdelnabi & Salem (2025), *The Hawthorne Effect in Reasoning Models*, arXiv:2505.14617 (NeurIPS). Cox, Kianersi & Garriga-Alonso (2026), *Decoding Answers Before Chain-of-Thought*, arXiv:2603.01437. Lanham et al. (2023), *Measuring Faithfulness in Chain-of-Thought Reasoning*. Lindsey (2025), *Emergent Introspective Awareness in Large Language Models*. Röttger et al. (2024), *XSTest*. Turpin et al. (2023), *LMs Don’t Always Say What They Think*.